Mathematical Modeling

Chapter 4: Degeneracy, Duality, and Dual Simplex

Personal study notes

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1 A common notation for both simplex methods

Consider the standard maximization problem

$$\max c^T x, \qquad Ax = b, \qquad x \geq 0, \qquad A \in \mathbb{R}^{m \times n}, \quad \operatorname{rank} A = m.$$

An ordered basis $B = (j_1, \ldots, j_m)$ selects a nonsingular A_B; N contains the nonbasic indices. Define

$$\bar{A} = A_B^{-1} A, \qquad \bar{b} = A_B^{-1} b, \qquad q = \bar{A}^T c_B - c, \qquad z_0 = c_B^T \bar{b}. \tag{1}$$

The basic solution has $x_N = 0$ and $x_B = \bar{b}$. Its equations and objective are

$$x_B = \bar{b} - \bar{A}_N x_N, \qquad z(x_N) = z_0 - \sum_{j \in N} q_j x_j. \tag{2}$$

Throughout these notes, $q_j = z_j - c_j$ is an *optimality indicator*. The more usual reduced cost is $\rho_j = -q_j$; never mix their signs. Row r corresponds to variable x_{j_r}, not necessarily x_r.

Condition	Meaning	Simplex role
$\bar{b} \geq 0$	Primal feasible basic solution	Primal method preserves it
$q \geq 0$	Dual feasible basis	Dual method preserves it
Both conditions	Optimal primal and dual pair	Stop with a certificate
$\bar{b}_i = 0$ for some i	Degenerate feasible basis	Zero steps may occur

A primal pivot chooses $k \in N$ with $q_k < 0$. For $u = \bar{A}_{:k}$, $x_k = \theta$ gives $x_B = \bar{b} - \theta u$ and objective change $-q_k \theta$. If some $u_i > 0$, the largest feasible step is

$$\theta^* = \min_{i : u_i > 0} \frac{\bar{b}_i}{u_i}.$$

If $u \leq 0$, the objective is unbounded above along this feasible ray. Both methods use the same Gauss–Jordan pivot operation. If $p = \bar{a}_{rk} \neq 0$,

$$\begin{aligned} \bar{a}'_{rj} &= \bar{a}_{rj}/p, & \bar{b}'_r &= \bar{b}_r/p, \\ \bar{a}'_{ij} &= \bar{a}_{ij} - \bar{a}_{ik}\bar{a}_{rj}/p, & \bar{b}'_i &= \bar{b}_i - \bar{a}_{ik}\bar{b}_r/p \quad (i \neq r), \\ q'_j &= q_j - q_k \bar{a}_{rj}/p, & z'_0 &= z_0 - q_k \bar{b}_r/p. \end{aligned}$$

2 Degeneracy: changing a basis without changing the point

A basic feasible solution is degenerate when at least one *basic* coordinate is zero. Having nonbasic zeros is normal and says nothing about degeneracy. Two mechanisms deserve separate attention.

Ratio test	Next point	Objective for $q_k < 0$
$\theta^* = 0$	Same full vector x; a different basis	No change
$\theta^* > 0$, unique minimum	A different point	Strict improvement
$\theta^* > 0$, tied minimum	A different, degenerate point: another basic variable also reaches zero	Strict improvement

For a tied positive ratio, $\bar{b}_i - \theta^* u_i = 0$ for every tied row. Only one variable leaves the basis; the others become zero basic variables. Thus degeneracy may be reached by a strictly improving step. A zero step need not prove that the current point is suboptimal: a degenerate optimal point can have a basis with some negative indicators.

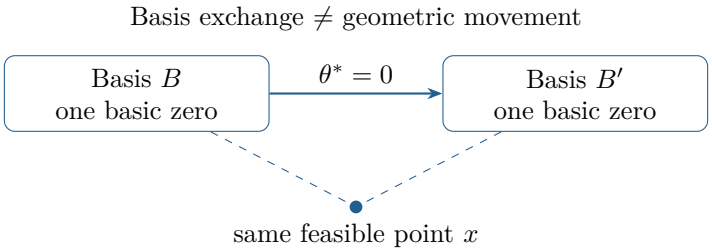


Figure 1: A degenerate pivot can change the equations used to describe a vertex without moving away from that vertex.

There are finitely many bases. If every pivot strictly improves the objective, no basis can recur. Zero steps remove this argument. With deterministic choices, a return to the same ordered basis repeats the same tableau and can initiate an endless cycle. Repeated points alone do not establish a cycle: several successive bases can describe the same point before the algorithm escapes.

2.1 A six-pivot cycling example

Consider seven nonnegative variables and the problem

$$\begin{aligned} \max \quad & \tfrac{3}{4}x_4 - 20x_5 + \tfrac{1}{2}x_6 - 6x_7, \\ & x_1 + \tfrac{1}{4}x_4 - 8x_5 - x_6 + 9x_7 = 0, \\ & x_2 + \tfrac{1}{2}x_4 - 12x_5 - \tfrac{1}{2}x_6 + 3x_7 = 0, \\ & x_3 + x_6 = 1, \qquad x_1, \ldots, x_7 \geq 0. \end{aligned} \tag{3}$$

Start at $B = (1, 2, 3)$, so $x = (0, 0, 1, 0, 0, 0, 0)^T$ and $z_0 = 0$. Use Dantzig's entering rule (most negative q_j) and resolve a tied ratio by taking the first row in the current tableau. The successive tableaus below show the complete cycle. The bottom row is q^T with z_0 on the right; basic columns are unit columns.

Basis $B = (1, 2, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_1	1	0	0	1/4	-8	-1	9	0
x_2	0	1	0	1/2	-12	$-1/2$	3	0
x_3	0	0	1	0	0	1	0	1
q^T	0	0	0	$-3/4$	20	$-1/2$	6	0

Basis $B = (4, 2, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	4	0	0	1	-32	-4	36	0
x_2	-2	1	0	0	4	3/2	-15	0
x_3	0	0	1	0	0	1	0	1
q^T	3	0	0	0	-4	$-7/2$	33	0

Basis $B = (4, 5, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	-12	8	0	1	0	8	-84	0
x_5	$-1/2$	$1/4$	0	0	1	$3/8$	$-15/4$	0
x_3	0	0	1	0	0	1	0	1
q^T	1	1	0	0	0	-2	18	0

Basis $B = (6, 5, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_6	$-3/2$	1	0	$1/8$	0	1	$-21/2$	0
x_5	$1/16$	$-1/8$	0	$-3/64$	1	0	$3/16$	0
x_3	$3/2$	-1	1	$-1/8$	0	0	$21/2$	1
q^T	-2	3	0	$1/4$	0	0	-3	0

Basis $B = (6, 7, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_6	2	-6	0	$-5/2$	56	1	0	0
x_7	$1/3$	$-2/3$	0	$-1/4$	$16/3$	0	1	0
x_3	-2	6	1	$5/2$	-56	0	0	1
q^T	-1	1	0	$-1/2$	16	0	0	0

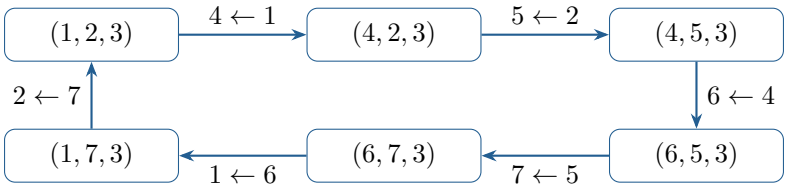
Basis $B = (1, 7, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_1	1	-3	0	$-5/4$	28	$1/2$	0	0
x_7	0	$1/3$	0	$1/6$	-4	$-1/6$	1	0
x_3	0	0	1	0	0	1	0	1
q^T	0	-2	0	$-7/4$	44	$1/2$	0	0

Basis $B = (1, 2, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_1	1	0	0	$1/4$	-8	-1	9	0
x_2	0	1	0	$1/2$	-12	$-1/2$	3	0
x_3	0	0	1	0	0	1	0	1
q^T	0	0	0	$-3/4$	20	$-1/2$	6	0

Every step has length zero. The first entry is x_4, with ratios $0/(1/4)$ and $0/(1/2)$; choosing row 1 makes x_1 leave. At the fourth tableau, $q_1 = -2$ and $q_7 = -3$, so Dantzig's rule chooses x_7. After six pivots, both the basis and the full tableau are back at their start.



Every pivot: $\theta^* = 0$, $x = (0, 0, 1, 0, 0, 0, 0)^T$, $z_0 = 0$

Figure 2: A cycle in the space of bases. Labels mean "entering index ← leaving index"; there is no corresponding tour of distinct vertices.

2.2 Bland's rule and its complete path

Bland's anti-cycling rule changes *both* choices:

1. Choose the smallest original nonbasic index k with $q_k < 0$.
2. Apply the ordinary minimum-ratio test; among ties, choose the smallest original index j_r of the basic variable that would leave.

In exact arithmetic this rule prevents cycling and yields finite termination of primal simplex. The claim is not a promise about speed or floating-point roundoff. A rule changing only the entering choice is not the full Bland rule.

Step	Ordered basis	Enter	Leave	θ^*	z_0
0	$(1, 2, 3)$	4	1	0	0
1	$(4, 2, 3)$	5	2	0	0
2	$(4, 5, 3)$	6	4	0	0
3	$(6, 5, 3)$	1	5	0	0
4	$(6, 1, 3)$	2	3	1/2	0
5	$(6, 1, 2)$	4	2	1	1/2
6	$(6, 1, 4)$	—	—	—	5/4

At step 3, Bland chooses x_1 rather than x_7, breaking the previous cycle. For that column the only positive transformed coefficient is $1/16$ in the x_5 row, so x_5 leaves at zero step. The remaining pivots move to

$$x^* = (3/4, 0, 0, 1, 0, 1, 0)^T, \qquad z^* = 5/4.$$

The final indicators are $q = (0, 3/2, 5/4, 0, 2, 0, 21/2)^T \geq 0$. Equation (2) therefore gives the exact certificate

$$z = \tfrac{5}{4} - \tfrac{3}{2}x_2 - \tfrac{5}{4}x_3 - 2x_5 - \tfrac{21}{2}x_7 \leq \tfrac{5}{4}. \tag{4}$$

All four nonbasic indicators are strictly positive. Equality forces their variables to zero, after which the basis equations fix the remaining variables; the optimum is unique.

Practice prompt. Rozwiązać przykład Beale, stosując regułę Bland'a. The path above supplies a hand solution; the MATLAB companion checks its choices, steps, and certificate.

3 Perturbation and lexicographic tie-breaking

Degeneracy also has a column-space interpretation. For a fixed basis,

$$b = \sum_{i=1}^{m} a_{j_i} \bar{b}_i.$$

A feasible basis writes b as a nonnegative combination of its columns. A degenerate basis places b on the boundary of that simplicial cone: at least one coefficient is zero. A small symbolic perturbation can place it in the interior of a nearby cone.

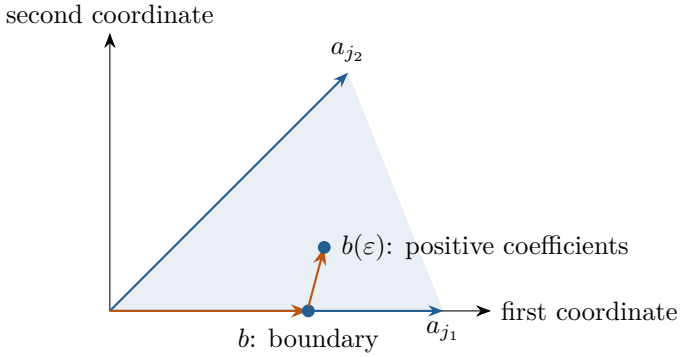


Figure 3: A two-dimensional illustration in the space of right-hand sides. The decision vector and the right-hand-side vector belong to different spaces.

3.1 Symbolic infinitesimals

Relabel columns so the initial feasible basis is $B_0 = (1, \ldots, m)$. Consider the formal perturbation

$$b(\varepsilon) = b + \sum_{j=1}^{n} \varepsilon^j a_j, \qquad 0 < \varepsilon \ll 1. \tag{5}$$

Here ε^2 is much smaller than ε, and so on. At the initial basis, the first m transformed columns are the identity. Consequently each zero basic coordinate gets a positive leading perturbation, so this basis stays feasible for sufficiently small positive ε. Without an initial ordering or a perturbation adapted to B_0, feasibility of the starting basis must be checked; arbitrary RHS noise is not this rule.

For any basis,

$$\bar{b}_i(\varepsilon) = \bar{b}_i + \sum_{j=1}^{n} \varepsilon^j \bar{a}_{ij}.$$

Every such polynomial is nonzero as a polynomial: a basic unit column supplies a nonzero coefficient. There are finitely many bases and rows, hence only finitely many positive roots. Choose ε below their smallest positive root, if there is one. Then no basic coordinate equals zero in the perturbed problem. Feasible pivots with $q_k < 0$ strictly improve its objective and cannot cycle. This is an exact-arithmetic argument, not a recommended numerical choice of a tiny floating-point constant.

When $\bar{a}_{ik} > 0$, compare the coefficient tuples

$$L_i = \left(\frac{\bar{b}_i}{\bar{a}_{ik}}, \frac{\bar{a}_{i1}}{\bar{a}_{ik}}, \ldots, \frac{\bar{a}_{in}}{\bar{a}_{ik}} \right) \tag{6}$$

lexicographically: compare the first coordinate, then the first coordinate where the tuples differ. Choose the smallest tuple. This implements the perturbed ratio test without assigning a numerical

value to ε. Distinct normalized rows cannot coincide, since $\bar{A}$ has full row rank. An equivalent shorter perturbation uses $b + A_{B_0}(\varepsilon, \ldots, \varepsilon^m)^T$ and keeps the transformed B_0 columns as its tie-break coefficients.

First tie in the cycling example. For $k = 4$, the ordinary ratio is zero in both first rows. Their tuples begin

$$L_1 = (0, 4, 0, 0, \ldots), \qquad L_2 = (0, 0, 2, 0, \ldots).$$

Since $0 < 4$ in the first differing position, row 2 wins: x_2 leaves. This gives the following two noninitial tableaus.

Basis $B = (1, 4, 3)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_1	1	$-1/2$	0	0	-2	$-3/4$	$15/2$	0
x_4	0	2	0	1	-24	-1	6	0
x_3	0	0	1	0	0	1	0	1
q^T	0	$3/2$	0	0	2	$-5/4$	$21/2$	0

Basis $B = (1, 4, 6)$.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_1	1	$-1/2$	$3/4$	0	-2	0	$15/2$	$3/4$
x_4	0	2	1	1	-24	0	6	1
x_6	0	0	1	0	0	1	0	1
q^T	0	$3/2$	$5/4$	0	2	0	$21/2$	$5/4$

After this zero step, x_6 enters. The only positive coefficient in that column is 1 in the x_3 row, so x_3 leaves with step 1. The optimizer is again $(3/4, 0, 0, 1, 0, 1, 0)^T$ and $z = 5/4$. The perturbed objectives strictly improve even when the unperturbed step is zero. As $\varepsilon \to 0$, an optimal certifying basis of the perturbed problem gives a feasible limit with unchanged nonnegative indicators, hence an optimum of the original problem.

Rule on this example	Pivots	Behavior	Optimum
Dantzig + first-row ties	6, then repeat	Cycles at the start	Not reached
Full Bland	6	Four zero steps, two moves	5/4
Dantzig + lexicographic ties	2	One zero step, one move	5/4

The counts compare this particular example, not universal relative performance.

4 Complexity and the route to an optimum

Avoiding cycles establishes termination, but does not establish a polynomial number of pivots. A family of distorted cubes exposes the difference. For $p = 5$, the three-dimensional member is

$$\begin{aligned} \max \quad & 4x_1 + 2x_2 + x_3, \\ & x_1 \le 5, \\ & 4x_1 + x_2 \le 25, \\ & 8x_1 + 4x_2 + x_3 \le 125, \qquad x \ge 0. \end{aligned}$$

Starting at the origin, Dantzig's rule visits all eight vertices:

Step	(x_1, x_2, x_3)	$4x_1 + 2x_2 + x_3$	Entering index next
0	$(0, 0, 0)$	0	1
1	$(5, 0, 0)$	20	2
2	$(5, 5, 0)$	30	4
3	$(0, 25, 0)$	50	3
4	$(0, 25, 25)$	75	1
5	$(5, 5, 65)$	95	5
6	$(5, 0, 85)$	105	4
7	$(0, 0, 125)$	125	—

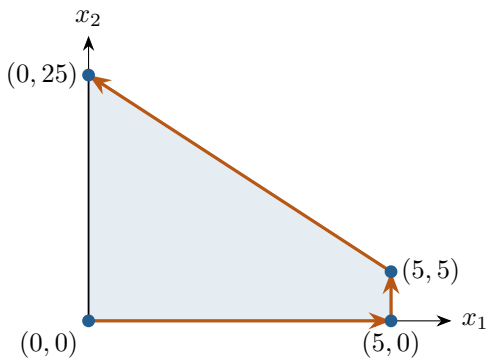
Indices 4–6 are the slack variables. The last constraint yields

$$4x_1 + 2x_2 + x_3 \le 125 - 4x_1 - 2x_2 \le 125,$$

with equality only at $(0, 0, 125)$. The analogous n-variable family maximizes $\sum_{j=1}^{n} 2^{n-j} x_j$, with rows

$$x_i + \sum_{j<i} 2^{i-j+1} x_j \le p^i, \quad i = 1, \ldots, n, \qquad x \ge 0.$$

For the usual choice $p = 5$, this rule can require $2^n - 1$ pivots. This is a worst-case statement about a specified pivot rule and family. It does not say that every simplex rule follows this path, or that simplex is slow on every LP.



Two-dimensional member
$\max(2x_1 + x_2)$
$x_1 \le 5,\ 4x_1 + x_2 \le 25$

Figure 4: The distorted square already makes Dantzig's rule visit every vertex. The higher-dimensional family extends this long edge-following route.

For rational input, polynomial-time complexity is measured against the *bit length* of all input coefficients, not just $m+n$. Ellipsoid and suitable interior-point algorithms have polynomial-time guarantees. Simplex is widely effective in practice, but no universal "average linear in $m + n$" bound follows without a stated probability model and analysis. Floating-point conditioning and sparse linear algebra also affect real runtime.

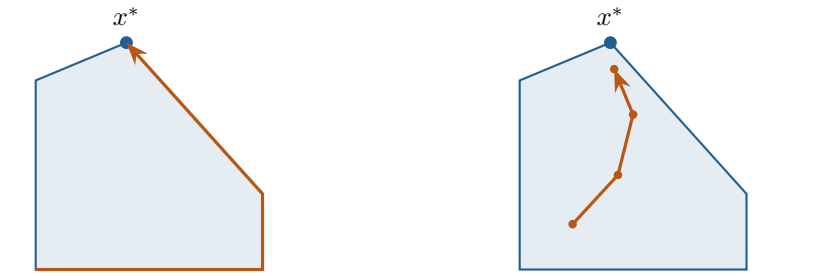


Figure 5: Schematic algorithmic routes, not computed iterates. An interior-point method approaches an optimal face; that face need not be a single vertex. Crossover can subsequently select a basic solution.

5 Constructing the dual problem

5.1 Bounds from weighted constraints

For the inequality-form primal

$$\text{(P)} \quad \max c^T x : \ Ax \le b, \ x \ge 0, \qquad \text{(D)} \quad \min b^T y : \ A^T y \ge c, \ y \ge 0, \tag{7}$$

think of y_i as the weight or price of resource constraint i. Nonnegative weights preserve its direction. If $A^T y \ge c$, each product's weighted resource use covers its objective coefficient, so $b^T y$ bounds the value achievable with available resources. Minimizing chooses the tightest bound.

The standard equality-form primal has dual

$$\max c^T x : \ Ax = b, \ x \ge 0 \quad \longleftrightarrow \quad \min b^T y : \ A^T y \ge c, \ y \in \mathbb{R}^m. \tag{8}$$

Equality constraints allow weights of either sign. Requiring $y \ge 0$ here would generally change the dual problem.

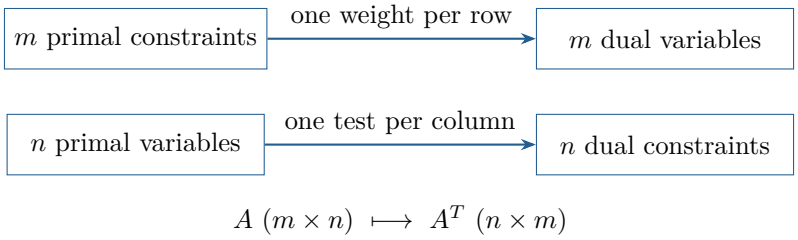


Figure 6: Dualization interchanges rows and columns, b and c, and max and min. The number of constraints changes places with the number of variables.

5.2 A sign table for mixed forms

Write the primal as a *maximization*. Then the dual is a minimization, with objective $b^T y$, and use the following two independent tables.

Primal row	Dual variable	Primal variable	Dual column condition
$a_i^T x \le b_i$	$y_i \ge 0$	$x_j \ge 0$	$a_j^T y \ge c_j$
$a_i^T x \ge b_i$	$y_i \le 0$	$x_j \le 0$	$a_j^T y \le c_j$
$a_i^T x = b_i$	y_i free	x_j free	$a_j^T y = c_j$

Here a_i^T denotes row i, while a_j denotes column j. For a primal *minimization*, reverse the inequality directions in the column conditions and the sign directions for inequality-row weights:

Primal row	Dual variable	Primal variable	Dual column condition
$a_i^T x \geq b_i$	$y_i \geq 0$	$x_j \geq 0$	$a_j^T y \leq c_j$
$a_i^T x \leq b_i$	$y_i \leq 0$	$x_j \leq 0$	$a_j^T y \geq c_j$
$a_i^T x = b_i$	y_i free	x_j free	$a_j^T y = c_j$

A quick derivation for max is to form

$$L(x, y) = c^T x + y^T(b - Ax) = b^T y + (c - A^T y)^T x.$$

The row sign rules ensure $L(x, y) \geq c^T x$ on primal-feasible x. The column rules ensure its supremum over the variable domains is $b^T y$: a positive coefficient on a nonnegative variable would otherwise give $+\infty$; a free variable requires a zero coefficient. Thus the signs follow from an upper-bound construction, rather than memorization alone. Dualizing again restores the original problem, up to equivalent sign conventions.

5.3 Worked mixed-sign example

$$\begin{array}{ll} \text{Primal: max} & 8x_1 + 3x_2 - 2x_3 \\ \text{subject to} & x_1 - 6x_2 + x_3 \geq 2, \\ & 5x_1 + 7x_2 - 2x_3 = -4, \\ & x_1 \leq 0, \quad x_2 \geq 0, \quad x_3 \in \mathbb{R}. \end{array}$$

The first row gives $y_1 \leq 0$; the second makes y_2 free. Read one column at a time to obtain

$$\begin{array}{ll} \text{Dual: min} & 2y_1 - 4y_2 \\ \text{subject to} & y_1 + 5y_2 \leq 8, \\ & -6y_1 + 7y_2 \geq 3, \\ & y_1 - 2y_2 = -2, \\ & y_1 \leq 0, \quad y_2 \in \mathbb{R}. \end{array}$$

As a check, $y = (0, 1)$ is dual feasible. The primal point $x = (0, 0, 2)$ is feasible and gives $c^T x = -4 = b^T y$. Equal feasible values certify that both points are optimal, as the next section proves.

6 Weak duality, strong duality, and possible outcomes

Theorem 6.1 (Weak duality)**.** *For the pair in* (7)*, every primal-feasible x and dual-feasible y satisfy $c^T x \leq b^T y$.*

Proof. Using nonnegativity in the two separate comparisons,

$$c^T x \leq x^T A^T y = y^T A x \leq y^T b.$$

□

Corollary 6.2 (Equal feasible values certify optimality)**.** *If feasible x, y satisfy $c^T x = b^T y$, both are optimal.*

Proof. Every feasible primal value is at most $b^T y = c^T x$, and every feasible dual value is at least $c^T x = b^T y$. □

This certificate uses only weak duality. The stronger theorem supplies the existence of a matching feasible bound.

Theorem 6.3 (Strong duality for linear programming)**.** *If one member of a primal–dual LP pair has a finite optimal value and is feasible, both attain finite optima and those values are equal.*

Proof through a certifying simplex basis. Convert the primal to full-row-rank standard form, removing redundant consistent equalities. Two-phase primal simplex with an anti-cycling rule terminates. Feasibility rules out failure in Phase I; a finite maximum rules out an improving unbounded ray. Its final basis has $\bar{b} \geq 0$ and $q \geq 0$. Define $y = A_B^{-T} c_B$. Then $A^T y - c = q \geq 0$, so y is feasible for the equality-form dual, and

$$b^T y = c_B^T A_B^{-1} b = c_B^T x_B = c^T x.$$

Weak duality certifies both optima. The equivalent transformations recover a dual certificate for the original form. Applying the same argument with the roles exchanged proves the assertion when the dual optimum is given first. □

An arbitrary basis representing a degenerate optimal point need not have $q \geq 0$. Use a *certifying* optimal basis when extracting y this way.

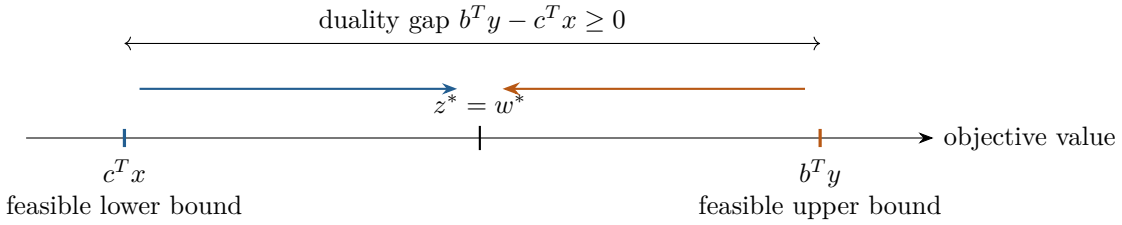


Figure 7: For a finite maximization LP, feasible primal values bound the optimum from below and feasible dual values bound it from above. Equality closes the gap.

6.1 The outcome table

"Unbounded" below means *feasible with an unbounded objective*, not an unbounded feasible set. The dual's unbounded direction is toward $-\infty$.

Primal \ dual	Finite optimum	Unbounded below	Infeasible
Finite optimum	Possible; equal values	Impossible	Impossible
Unbounded above	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

An improving primal ray contradicts any finite feasible dual upper bound. So primal unboundedness implies dual infeasibility. The converse does not follow: both problems can be infeasible.

Case	Primal	Dual
Finite / finite	$\max x$, $0 \leq x \leq 1$	$\min y$, $y \geq 1$
Unbounded / infeasible	$\max x$, $0x \leq 1$, $x \geq 0$	$\min y$, $0y \geq 1$, $y \geq 0$
Infeasible / unbounded	$\max 0x$, $x \leq -1$, $x \geq 0$	$\min(-y)$, $y \geq 0$
Infeasible / infeasible	$\max x$, $0x \leq -1$, $x \geq 0$	$\min(-y)$, $0y \geq 1$, $y \geq 0$

The zero rows make the last examples deliberately simple; no full-rank standard-form assumption is needed to state these duality outcomes.

7 Reading a dual certificate from an optimal tableau

Return to the production model

$$\max(3x_1 + 2x_2), \qquad \begin{cases} 2x_1 + x_2 \le 10, \\ 3x_1 + 3x_2 \le 24, \\ \frac{3}{2}x_1 \le 6, \\ x_1, x_2 \ge 0. \end{cases}$$

Introduce slacks x_3, x_4, x_5 in that row order. The final primal tableau is

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_2	0	1	-1	$2/3$	0	6
x_5	0	0	$-3/2$	$1/2$	1	3
x_1	1	0	1	$-1/3$	0	2
q^T	0	0	1	$1/3$	0	18

Its basis is $B = (2, 5, 1)$; $c_B = (2, 0, 3)^T$. Solve $A_B^T y = c_B$, rather than forming an explicit inverse:

$$A_B = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 0 & 3 \\ 0 & 1 & 3/2 \end{pmatrix}, \qquad \begin{cases} y_1 + 3y_2 = 2, \\ y_3 = 0, \\ 2y_1 + 3y_2 + \frac{3}{2}y_3 = 3. \end{cases}$$

Thus $y = (1, 1/3, 0)^T$. It is feasible for the dual

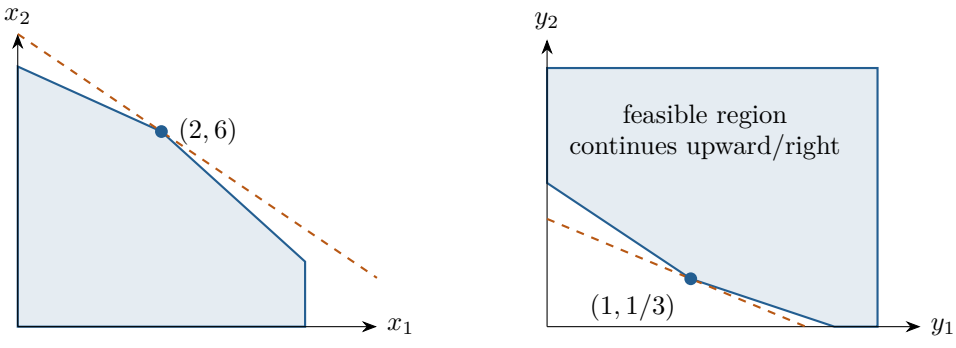
$$\min(10y_1 + 24y_2 + 6y_3), \quad 2y_1 + 3y_2 + \tfrac{3}{2}y_3 \ge 3, \quad y_1 + 3y_2 \ge 2, \quad y \ge 0,$$

and $b^T y = 10 + 8 = 18 = c^T x^*$ for $x^* = (2, 6)^T$.

More generally, for $[A \mid I]x = b$, $c_{\text{std}} = (c, 0)$,

$$q_{1:n} = A^T y - c, \qquad q_{n+1:n+m} = y. \tag{9}$$

The indicators under the primal slack columns are the dual variables. The indicators under the original columns are the dual constraint surpluses. This reading rule depends on the actual $+I$ slack convention; do not carry it unchanged to surplus columns or rows multiplied by -1.



Primal: $\max(3x_1 + 2x_2)$ Dual slice: $y_3 = 0$

Figure 8: The primal optimum and the dual certificate both have objective 18. The right drawing is the explicitly stated slice $y_3 = 0$, not a projection of the complete three-dimensional dual region. Dashed lines are objective level sets.

7.1 Resource prices and their range of validity

The identity

$$3x_1 + 2x_2 = (2x_1 + x_2) + \tfrac{1}{3}(3x_1 + 3x_2)$$

prices the first two resources at 1 and 1/3 PLN per resource unit; the third has price zero at this basis. For a change $b \mapsto b + \delta$, with A, c fixed, the same basis remains optimal whenever $A_B^{-1}(b + \delta) \geq 0$. In that range,

$$z^*(b + \delta) = 18 + y^T \delta.$$

This equality has a feasibility condition, not global validity for arbitrary resource changes. For example, changing only the first RHS by t gives

$$(x_2, x_5, x_1) = (6 - t,\ 3 - \tfrac{3}{2}t,\ 2 + t).$$

The basis stays feasible exactly for $-2 \leq t \leq 2$, and revenue is $18 + t$.

t	x_1	x_2	Third slack	Revenue (PLN)
-2	0	8	6	16
0	2	6	3	18
2	4	4	0	20

At the endpoints the basis becomes degenerate; outside this interval, recompute the optimum. A zero price and a tight constraint can coexist at degeneracy.

8 Complementary slackness

For the max/min pair (7), take feasible x, y and set

$$s = b - Ax \geq 0, \qquad t = A^T y - c \geq 0.$$

Their duality gap splits into nonnegative products:

$$b^T y - c^T x = y^T(b - Ax) + x^T(A^T y - c) = y^T s + x^T t. \tag{10}$$

Theorem 8.1 (Complementary slackness)**.** *A primal-feasible x and dual-feasible y are both optimal if and only if*

$$y_i s_i = 0 \quad (i = 1, \ldots, m), \qquad x_j t_j = 0 \quad (j = 1, \ldots, n).$$

Proof. For optimal feasible points, strong duality gives zero gap. Since every summand in (10) is nonnegative, each product must vanish. Conversely, vanishing products give equal feasible objectives; weak duality certifies optimality. □

The points do *not* need to be basic. Feasibility is essential: zero products alone cannot certify an inadmissible point.

Positive quantity	Forced zero quantity
Primal slack $s_i > 0$	Corresponding price $y_i = 0$
Price $y_i > 0$	Corresponding slack $s_i = 0$
Primal variable $x_j > 0$	Dual surplus $t_j = 0$
Dual surplus $t_j > 0$	Primal variable $x_j = 0$

These are implications, not equivalences. Both members of a complementary pair can equal zero. In the production example, $s = (0, 0, 3)$, $y = (1, 1/3, 0)$, $x = (2, 6)$, and $t = (0, 0)$, so every product vanishes.

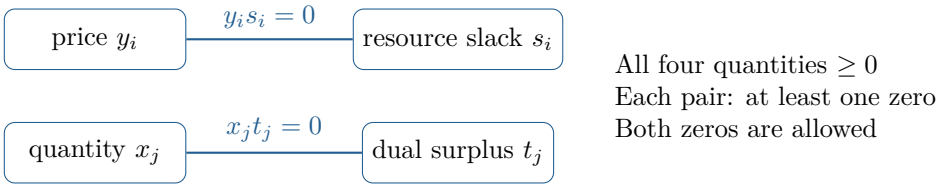


Figure 9: Complementarity pairs each resource with its price, and each decision variable with the gap in its dual constraint.

8.1 Certifying a supplied candidate without running simplex

Task statement. Sprawdzić, czy $x = [2, 1, 1, 0, 0]^T$ jest rozwiązaniem optymalnym. Consider

$$\begin{aligned} \min \quad & x_1 + 4x_2 + 2x_3 + 2x_4 + 8x_5, \\ & x_1 + 2x_2 + x_4 \geq 4, \\ & x_2 + x_3 - x_4 \geq 2, \\ & x_2 + x_5 \geq 1, \qquad x_1, \ldots, x_5 \geq 0. \end{aligned}$$

For a min problem $Ax \geq b$, the dual is

$$\max b^T y, \quad A^T y \leq c, \quad y \geq 0.$$

Writing it explicitly gives

$$\begin{aligned} \max \quad & 4y_1 + 2y_2 + y_3, \\ & y_1 \leq 1, \qquad 2y_1 + y_2 + y_3 \leq 4, \qquad y_2 \leq 2, \\ & y_1 - y_2 \leq 2, \qquad y_3 \leq 8, \qquad y_1, y_2, y_3 \geq 0. \end{aligned}$$

The candidate is nonnegative and has $Ax = (4, 2, 1)^T = b$. Since its first three coordinates are positive, complementary slackness forces their dual constraints to be tight:

$$y_1 = 1, \quad 2y_1 + y_2 + y_3 = 4, \quad y_2 = 2.$$

Hence $y = (1, 2, 0)^T$. The remaining inequalities give $-1 \leq 2$ and $0 \leq 8$, so it is dual feasible. The min-form gap uses $s = Ax - b$ and $t = c - A^T y$:

$$c^T x - b^T y = y^T s + x^T t.$$

j	x_j	$t_j = c_j - a_j^T y$	$x_j t_j$
1	2	0	0
2	1	0	0
3	1	0	0
4	0	3	0
5	0	8	0

All three primal surpluses are zero, so $y_i s_i = 0$ also holds. The objectives match: $c^T x = 2+4+2 = 8 = 4 + 4 = b^T y$. Thus both candidates are optimal. Moreover, positive dual surpluses force $x_4 = x_5 = 0$, and $y_1, y_2 > 0$ force the first two primal rows tight. Every primal optimum therefore has

$$x(t) = (4 - 2t,\ t,\ 2 - t,\ 0,\ 0)^T, \qquad 1 \leq t \leq 2. \tag{11}$$

The supplied candidate is the endpoint $t = 1$, not the only optimizer. The third row gives $t \geq 1$; nonnegativity gives $t \leq 2$. The dual certificate is unique: any dual optimum must be complementary to the supplied candidate, whose first three positive coordinates force $y = (1, 2, 0)$.

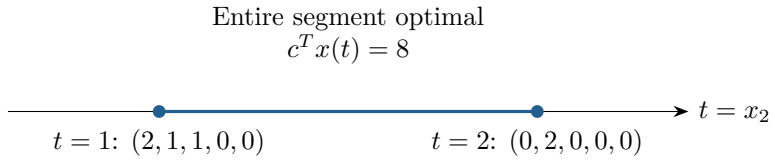


Figure 10: The parameter line represents a one-dimensional optimal face in the five-dimensional decision space. Interior points of this segment are also optimal.

9 The dual simplex method

For the standard max problem, a *dual feasible basis* satisfies $q \geq 0$; its basic solution may have negative coordinates. The associated $y = A_B^{-T} c_B$ satisfies $A^T y \geq c$, and $b^T y = z_0$, even if the primal basic vector is infeasible. A negative z_0 or $\bar{b}$ is not by itself an error. If also $\bar{b} \geq 0$, primal and dual feasibility and equal values certify optimality.

Feature	Primal simplex	Dual simplex
Maintained condition	$\bar{b} \geq 0$	$q \geq 0$
Condition to repair	Some $q_k < 0$	Some $\bar{b}_r < 0$
First selection	Entering column	Leaving row
Eligible pivot sign	$\bar{a}_{rk} > 0$	$\bar{a}_{rk} < 0$
Objective z_0 for max	Nondecreasing feasible value	Nonincreasing dual upper bound
Failure certificate	Improving primal ray	Contradictory primal row

The dual method is particularly convenient after changing b: indicators depend on A, c, B, so they are unchanged, while $\bar{b}$ may lose nonnegativity. It also naturally starts some min problems with nonnegative cost coefficients.

9.1 The reversed ratio test

Choose a row r with $\bar{b}_r < 0$, for instance the most negative RHS. Its equation is

$$x_{j_r} + \sum_{j \in N} \bar{a}_{rj} x_j = \bar{b}_r < 0.$$

If every nonbasic coefficient in that row is nonnegative, the left side cannot be negative for any nonnegative vector. The primal is infeasible. Otherwise a pivot can use a column with $\bar{a}_{rk} < 0$. Define $\alpha = q_k/(-\bar{a}_{rk}) \geq 0$. The new indicators are

$$q'_j = q_j + \alpha \bar{a}_{rj}.$$

For $\bar{a}_{rj} \geq 0$, nonnegativity is automatic. For negative coefficients, we require $\alpha \leq q_j/(-\bar{a}_{rj})$. Therefore choose

$$k \in \arg\min_{j:\bar{a}_{rj}<0} \frac{q_j}{-\bar{a}_{rj}} \quad \Longleftrightarrow \quad k \in \arg\max_{j:\bar{a}_{rj}<0} \frac{q_j}{\bar{a}_{rj}}. \tag{12}$$

This preserves dual feasibility. Since basic columns are unit columns, eligible columns are automatically nonbasic. The new entering basic value $\bar{b}_r/\bar{a}_{rk}$ is positive, but other basic values may still be negative. Continue until all are nonnegative.

The objective update is

$$z'_0 = z_0 + \alpha \bar{b}_r \leq z_0.$$

For a maximization problem this decreases a feasible *dual upper bound*, not a feasible primal objective. If $q_k = 0$, the dual step is degenerate and this bound need not strictly decrease.

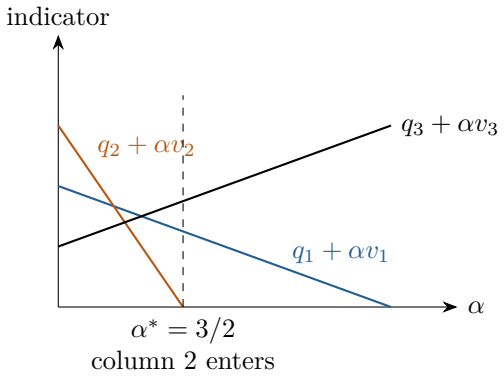


Figure 11: Added illustration: $q = (2, 3, 1)$ and pivot-row coefficients $v = (-1/2, -2, 1/2)$. The first indicator to reach zero limits the dual step: $\min(2/(1/2), 3/2) = 3/2$. The positive-coefficient column imposes no upper limit.

9.2 Algorithm and infeasibility certificate

1. Start with a nonsingular basis having $q \geq 0$.
2. If $\bar{b} \geq 0$, return the optimal primal basic vector and its dual certificate.
3. Choose a row r with negative RHS.
4. If that row has no negative nonbasic coefficient, certify infeasibility.
5. Otherwise use (12) and pivot on its negative element.
6. Recompute or update the tableau and return to step 2.

A smallest-column tie break makes a worked path reproducible, but the simple "most negative RHS" rule alone is not an anti-cycling theorem. General dual simplex needs a proven anti-cycling variant when degeneracy can occur.

To express failure without a tableau, put $h = A_B^{-T} e_r$. Then the contradictory row satisfies

$$h^T A = \bar{A}_{r:} \geq 0, \qquad h^T b = \bar{b}_r < 0.$$

Any $x \geq 0$ with $Ax = b$ would give $h^T b = h^T A x \geq 0$, a contradiction. This is an explicit infeasibility certificate. Starting at any dual-feasible y, the ray $y + \lambda h$ remains dual feasible and its objective tends to $-\infty$.

For example, $x_1 + x_2 = -1$, $x \geq 0$, $\max 0$ has the dual-feasible basis $B = (1)$ with $q = (0, 0)$ and RHS -1. Its row has no negative coefficient. The certificate is $h = 1$, and the dual $\min(-y)$, $y \geq 0$ is unbounded below. This single row simultaneously explains the two outcomes.

10 A complete dual-simplex solve

Consider

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3, \\ & x_1 + 2x_2 + x_3 \geq 2, \\ & 2x_1 + x_2 - x_3 \geq 10, \qquad x_1, x_2, x_3 \geq 0. \end{aligned}$$

For a general min problem $\min c^T x$, $Ax \geq b$, $x \geq 0$ with $c \geq 0$, negate the objective and rows and add surplus variables:

$$\max(-c^T x), \qquad -Ax + s = -b, \qquad x, s \geq 0.$$

The surplus basis is I, with $x = 0$, $s = -b$ and indicators $(c, 0) \geq 0$. It is dual feasible regardless of the signs of b, and primal feasible only when $-b \geq 0$. Thus no artificial variables are needed for this start.

Here x_4, x_5 denote the nonnegative surpluses of the original inequalities:

$$-x_1 - 2x_2 - x_3 + x_4 = -2, \qquad -2x_1 - x_2 + x_3 + x_5 = -10.$$

The start $B = (4, 5)$ is primal infeasible but dual feasible:

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_4	-1	-2	-1	1	0	-2
x_5	-2	-1	1	0	1	-10
q^T	1	1	1	0	0	0

The most negative RHS is in row 2, so x_5 leaves. Only columns 1 and 2 have negative entries there:

Column	q_j	$\bar{a}_{2j}$	$q_j/(-\bar{a}_{2j})$
1	1	-2	1/2
2	1	-1	1

Choose x_1 to enter, and pivot on -2. Divide row 2 by -2, add that normalized row to row 1, and subtract it from the indicator row:

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_4	0	$-3/2$	$-3/2$	1	$-1/2$	3
x_1	1	1/2	$-1/2$	0	$-1/2$	5
q^T	0	1/2	3/2	0	1/2	-5

Both RHS entries are now nonnegative and all indicators remain nonnegative. Thus $(x_1, x_2, x_3) = (5, 0, 0)$, surpluses $(3, 0)$, and the *original minimum* is 5. The transformed maximum is -5.

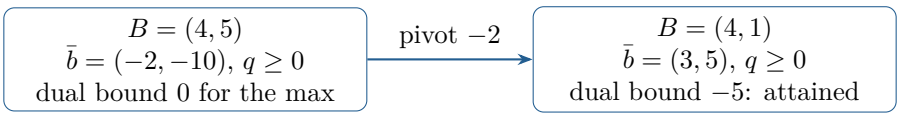
The original min-form dual is

$$\max(2y_1 + 10y_2), \quad y_1 + 2y_2 \leq 1, \quad 2y_1 + y_2 \leq 1, \quad y_1 - y_2 \leq 1, \quad y \geq 0.$$

It has certificate $y = (0, 1/2)$, objective 5, and column slacks $(0, 1/2, 3/2)$. Equivalently, solve $A_B^T y = c_B$ for the transformed max to get the same nonnegative y: its surplus columns form $+I$ after row negation. The original primal inequalities directly imply the bound

$$x_1 + x_2 + x_3 = 5 + \tfrac{1}{2}x_2 + \tfrac{3}{2}x_3 + \tfrac{1}{2}x_5 \geq 5. \tag{13}$$

Equality forces $x_2 = x_3 = x_5 = 0$, then $x_1 = 5$; the optimum is unique.



Original min: lower bound rises from 0 to 5
Primal feasibility is repaired in one pivot

Figure 12: Dual simplex keeps a valid dual certificate throughout. The starting basic vector is not an admissible solution of the original minimization problem.

11 MATLAB checks and study questions

The companion uses base MATLAB and the same matrices as the hand examples:

```
run('matlab/exercises/lecture_04_checks.m')
```

Basis computations use linear solves, not an explicit inverse:

```
AB = A(:, B);
Abar = AB \ A;
bbar = AB \ b;
y = AB' \ c(B);
q = A' * y - c;
x = zeros(size(A, 2), 1);
x(B) = bbar;
```

The assertions check specified small examples, not a general-purpose LP solver. They compare known paths, feasible points, certificates, and sign conditions with absolute tolerance 10^{-10}. Lexicographic comparisons use the same tolerance for these moderate-size examples; this is not exact symbolic arithmetic or a universal anti-cycling guarantee for ill-conditioned data.

Check	Expected hand result
Cycling example	Six zero steps; starting basis repeats
Bland / lexicographic paths	Six / two pivots; both attain $5/4$
Three-dimensional distorted cube	Eight vertices, seven pivots, maximum 125
Production dual and RHS changes	$y = (1, 1/3, 0)$; $18 + t$ for $-2 \le t \le 2$
Complementary-slackness candidate	Value 8; entire optimal segment $1 \le t \le 2$
Dual-simplex solve	One pivot; minimum 5; $y = (0, 1/2)$
Contradictory tableau row	$h^T A \ge 0$, $h^T b < 0$: infeasible

11.1 Common mistakes

Mistake	Repair
Calling every zero coordinate degenerate	Check the basic coordinates
Equating degeneracy with cycling	Distinguish a zero step from a repeated basis
Using the max sign table for a min primal	Reverse the appropriate signs, or negate the objective first
Treating a tight row as a positive-price guarantee	Complementarity gives implications; both quantities can be zero
Checking only equality of objective values	Verify feasibility on both sides
Reading all indicator rows as resource prices	Identify the actual original slack columns and row signs
Using the primal ratio test in dual simplex	Choose a negative RHS row first, then minimize $q_j/(-\bar{a}_{rj})$
Reporting transformed value -5 as the minimum	Undo objective negation: the minimum is 5

11.2 Questions with answers for checking

1. In the cycling example's fourth tableau, why do Dantzig and Bland choose different entering variables?
2. Why does the perturbation rule select row 2 in the first ratio tie?
3. For $\max 2x$, $x = 3$, $x \geq 0$, construct the dual and its optimum.
4. Does a zero resource price prove a resource is unused?
5. Derive every optimal primal point for the complementary-slackness task.
6. In the dual-simplex example, what happens if column 2 is chosen at the first pivot instead of column 1?

Answers. (1) Dantzig chooses $q_7 = -3$, while Bland chooses the smaller eligible index 1 with $q_1 = -2$. (2) The normalized tuples start $(0, 4, 0, \ldots)$ and $(0, 0, 2, \ldots)$. (3) $\min 3y$, $y \geq 2$, y free; optimum $y = 2$, value 6. (4) No: $y_i = 0$ permits either zero or positive slack. (5) Equation (11) follows from $x_4 = x_5 = 0$ and tight first two rows; all its points have value 8. (6) Its ratio 1 exceeds the allowed $1/2$; the updated $q_1 = 1 - 2 = -1$ loses dual feasibility. An arbitrary negative pivot is insufficient.

12 Sources and editorial remarks

Primary source: E. Pawelec, `s4_w4.pdf`, PDF pages 1–12, printed slides 1–47 (slide 47 is the closing slide).
Source date: The folder is labelled 2020; PDF creation metadata is 2019-10-27, not an established delivery date.
Current course alignment: Supplied sheet 1, groups 2–5, requires dual construction, complementarity and dual simplex. These topics are not developed in the supplied `ProgMat_w1.pdf`; its numbering does not establish when they were taught. Cycling and complexity remain historical enrichment.
Editorial scope: Complete mathematical lecture content, reorganized in English, with hand proofs, worked paths, certificates, study questions, and editable TikZ illustrations. All source pages were reviewed; dense cycling, perturbation, and final pivot tableaus were also inspected at enlarged scale.

PDF pages	Printed slides	Developed material
1	1–4	Section 1: standard form and pivot equations
2	5–8	Section 2: degeneracy and full cycling tableaus
3–4	9–14	Sections 3, 2.2: perturbation and Bland
4–5	15–18	Section 4: complexity and geometric routes
5–6	19–23	Section 5: dual construction and mixed signs
6–7	24–28	Section 6: weak/strong duality and outcomes
7–8	29–31	Section 7: extracting the production dual
8–9	32–36	Section 8: optimality via complementarity
10–11	37–44	Section 9: dual simplex and its ratio rule
12	45–46	Section 10: initialization and complete solve

The Polish practice prompt in Section 2.2 is from slide 14. The candidate task in Section 8.1 preserves the mathematical question on slide 34 with spelling normalized; its data are from slides 34–36. The full Bland path, the $p = 5$ three-dimensional vertex list, mixed-example feasible pair, resource-price interval, optimal segment, and explicit row infeasibility certificate are added derivations. Drawings explain the source ideas in the current style; the ratio-indicator graph uses added data.

Qualifications. Section 2 distinguishes zero steps from positive tied steps, and allows a negative-indicator basis at a degenerate optimal point. Section 3 makes the initial feasibility requirement explicit. Section 4 attaches the distorted-cube route to its pivot rule and does not adopt slide 16's unspecified average-complexity expression. Section 6 distinguishes the equal-value certificate on slide 25 from strong duality's existence statement. A certifying basis is needed to extract dual multipliers from an optimal point. Complementarity applies to feasible points, including nonbasic ones. Section 9.2 qualifies termination under dual degeneracy.

Supplementary mathematical checks. MIT OCW 15.081J/6.251J, Fall 2009: Lecture 6, The Simplex Method II, PDF pp. 5–8, for symbolic perturbation and anti-cycling; Lecture 8, Duality Theory I, PDF pp. 2–5, for dual signs, the strong-duality basis proof, and complementarity; Lecture 9, Duality Theory II, PDF pp. 4–5, for dual-simplex invariants and the ratio test. Titles here follow the PDF interiors, which differ from some catalogue labels. These references were consulted through the web tool; shell downloads failed at DNS resolution. Intended storage is outside this repository, under `../sources/mit/`; the paired source inventory records URLs and status. They do not establish current course coverage.

Verification. Exact rational calculations independently recomputed the cycling, Bland, lexicographic, distorted-cube, and dual-simplex paths and their certificates. The installed R2026b static analyzer reported no diagnostics. On 2026-10-04, the user ran the MATLAB companion in a normal terminal; all numerical assertions passed and MATLAB reported version 26.2.0.3386108 (R2026b). The supplied output agrees with Section 11. Execution evidence is recorded in `docs/matlab-verification.md`; Polish terms are in `docs/glossary.md`.